

CS3300 - GCP Deployment from a Packaged JAR

Complementary Student Notes | Build JAR -> App Engine

Use this workflow when your Spring Boot application has already been packaged as a JAR. This is a deployment-oriented workflow; the application is still deployed to App Engine.

1. Start with the Packaged Application

From the root of your Spring Boot project, build the application:

```
mvn clean package
```

Confirm that a JAR was created in the target directory:

```
dir target
```

Key idea: The exact JAR filename comes from the project artifactId and version in pom.xml. Do not assume a fixed filename.

2. Verify Google Cloud CLI

```
gcloud --version  
gcloud config get-value project
```

Authenticate/select the project if necessary:

```
gcloud init
```

3. Create or Confirm app.yaml

```
runtime: java21  
env: standard  
service: default
```

Place app.yaml in the deployment project directory.

4. Deploy the Packaged Application

Make sure the terminal is in the directory containing app.yaml, then run:

```
gcloud app deploy
```

Read the displayed project/application/region details before confirming.

5. Verify the Deployment

```
gcloud app browse
```

Open the appropriate application endpoint, for example:

```
https://<app-engine-url>/students
```

6. Workflow Summary

Source code -> Maven build -> target/<application>.jar + app.yaml -> gcloud app deploy -> App Engine

Key idea: Maven is responsible for building and packaging. App Engine is the cloud runtime. gcloud is the deployment/management interface.

7. When This Workflow Is Useful

- Your application has already been built and tested.
- You want to deploy a known packaged artifact.
- The source code does not need to be edited during the deployment step.